

Practical - Marking Rubric (20 marks)

Shared Mock Helpers (2 marks)

Mock Helper Files (1 mark)

Band	Marks	Criteria
A	0.8-1	All five mock helper files created (<code>category</code> , <code>quiz</code> , <code>question</code> , <code>gameSession</code> , <code>answer</code>). Each file correctly exports <code>mockReq</code> , <code>mockRes</code> and a repository stub factory for its domain. Stub factories cover all five CRUD methods. Helpers are imported into test files rather than redefined inline.
B	0.7	Four or five mock helper files present with mostly correct structure. Minor issues such as a missing stub method or a helper redefined in a test file rather than imported.
C	0.5-0.6	Some mock helper files present but incomplete or inconsistent. Some domains missing or stub factories only partially implemented.
D/E	0-0.4	Mock helper files largely absent or incorrectly structured. Helpers duplicated across test files rather than shared.

`mockRes` Chaining (1 mark)

Band	Marks	Criteria
A	0.8-1	<code>res.status</code> is implemented as a Sinon stub that returns <code>res</code> in all mock helper files. Chained calls such as <code>res.status(201).json(...)</code> work correctly in all test suites.
B	0.7	<code>res.status</code> returns <code>res</code> in most mock files. One or two files may have an implementation that does not support chaining correctly.
C	0.5-0.6	<code>mockRes</code> is implemented in some files but does not consistently support chaining. Tests may fail or require workarounds.
D/E	0-0.4	<code>mockRes</code> does not implement chaining. <code>res.status</code> does not return <code>res</code> . Tests relying on chained calls will fail.

Auth Controller Tests (4 marks)

Register - Success (1 mark)

Band	Marks	Criteria
A	0.8-1	Test correctly stubs the repository to simulate a new user. Asserts <code>201</code> status. Asserts the response body does not include a <code>password</code> field. <code>sinon.restore()</code> called in <code>afterEach</code> .

Band	Marks	Criteria
B	0.7	Test stubs the repository and asserts 201 . Password field check may be missing or incomplete. <code>sinon.restore()</code> present.
C	0.5-0.6	Test present but assertions incomplete. May assert status only or stub behaviour is incorrect.
D/E	0-0.4	Test missing or does not meaningfully verify register success behaviour.

Register - Duplicate User (1 mark)

Band	Marks	Criteria
A	0.8-1	Test correctly stubs the repository to simulate a user that already exists. Asserts 409 status with a meaningful error message in the response body.
B	0.7	Test stubs the repository and asserts 409 . Response body assertion may be missing.
C	0.5-0.6	Test present but stub behaviour does not correctly simulate a duplicate or wrong status code asserted.
D/E	0-0.4	Test missing or does not verify duplicate user behaviour.

Login - Success (1 mark)

Band	Marks	Criteria
A	0.8-1	Test correctly stubs the repository to return a valid user with a hashed password and stubs the password comparison to return true . Asserts 200 status and that the response body includes a token field.
B	0.7	Test asserts 200 and token presence. Password comparison stub may be missing or not fully realistic.
C	0.5-0.6	Test present but token assertion missing or stub setup incomplete.
D/E	0-0.4	Test missing or does not verify login success.

Login - Invalid Credentials (1 mark)

Band	Marks	Criteria
A	0.8-1	Test covers both sub-cases: user not found (repository returns null) and password mismatch (comparison returns false). Both assert 401 status.
B	0.7	Test covers at least one sub-case and asserts 401 . Second sub-case may be missing.
C	0.5-0.6	Test present but only one sub-case tested and assertion may be on the wrong status code.
D/E	0-0.4	Test missing or does not verify invalid credential behaviour.

Category Controller Tests (2 marks)

Band	Marks	Criteria
A	1.6-2	All five CRUD operations tested. Create asserts <code>201</code> . Read all tests both a non-empty result (<code>200</code>) and an empty result (<code>404</code>). Read by ID tests both found (<code>200</code>) and not found (<code>404</code>). Update and delete each test both the success and not-found paths. <code>afterEach(() => sinon.restore())</code> present. Shared helpers imported from <code>tests/mocks/category.mock.js</code> .
B	1.3-1.5	All five operations covered with minor omissions. One or two sub-cases (e.g. the empty array path or a not-found path) may be missing. Shared helpers used.
C	1-1.2	Basic coverage of some operations. Several sub-cases missing. Happy paths only or stubs not correctly simulating repository behaviour.
D/E	0-0.9	Few or no meaningful tests. Operations largely untested or tests do not correctly verify controller behaviour.

Quiz Controller Tests (2 marks)

Band	Marks	Criteria
A	1.6-2	All five CRUD operations tested with both success and failure paths, matching the depth of the category tests. Quiz-category relationship handled correctly in tests without requiring a real category. <code>afterEach(() => sinon.restore())</code> present. Shared helpers imported from <code>tests/mocks/quiz.mock.js</code> .
B	1.3-1.5	All five operations covered with minor omissions. Relationship handled adequately. Shared helpers used.
C	1-1.2	Basic operation coverage. Several sub-cases or the relationship scenario missing.
D/E	0-0.9	Few or no meaningful tests. Operations largely untested.

Question Controller Tests (2 marks)

CRUD Operations (1 mark)

Band	Marks	Criteria
A	0.8-1	All five CRUD operations tested with success and failure paths. Shared helpers imported from <code>tests/mocks/question.mock.js</code> . <code>afterEach(() => sinon.restore())</code> present.
B	0.7	All five operations covered with minor omissions.
C	0.5-0.6	Some operations covered but multiple sub-cases missing.
D/E	0-0.4	Few or no meaningful CRUD tests for questions.

OpenTDB Import Endpoint (1 mark)

Band	Marks	Criteria
A	0.8-1	Both sub-cases tested. Success case stubs <code>fetch</code> to return a valid OpenTDB response (<code>response_code: 0</code>) with at least one question, asserts <code>201</code> and a question count in the response body. Failure case stubs <code>fetch</code> to return a non-zero <code>response_code</code> and asserts an appropriate error status.
B	0.7	Both sub-cases present with minor issues. Success case may not assert the question count. Failure case may use an incorrect status code.
C	0.5-0.6	One sub-case tested or <code>fetch</code> not correctly stubbed. Assertions incomplete.
D/E	0-0.4	OpenTDB import not tested or test does not meaningfully verify the endpoint's behaviour.

Game Session Controller Tests (2 marks)

Start Session (1 mark)

Band	Marks	Criteria
A	0.8-1	Success case stubs both the quiz and player repositories to simulate valid records, asserts <code>201</code> and that the new session is in the response body. Not-found case stubs the quiz repository to return <code>null</code> and asserts <code>404</code> .
B	0.7	Both sub-cases present with minor issues. One assertion may be missing.
C	0.5-0.6	One sub-case tested. Stubs may not correctly reflect the controller's dependencies.
D/E	0-0.4	Start session not meaningfully tested.

Finalise Session (1 mark)

Band	Marks	Criteria
A	0.8-1	Success case stubs answers and verifies the score percentage is correctly calculated and present in the response body, asserts <code>200</code> . Error case stubs the session repository to return <code>null</code> or an already-finalised session and asserts an appropriate error status.
B	0.7	Both sub-cases present. Score calculation assertion may be missing or imprecise.
C	0.5-0.6	One sub-case tested or score calculation not verified.
D/E	0-0.4	Finalise session not meaningfully tested.

Answer Controller Tests (2 marks)

Submit Answer - Success (1 mark)

Band	Marks	Criteria
A	0.8-1	Test stubs both the game session and question repositories to return valid records. Asserts <code>201</code> . Asserts the response body includes a field indicating whether the answer was correct.
B	0.7	Test asserts <code>201</code> . Correctness field assertion may be missing.
C	0.5-0.6	Test present but stubs incomplete or assertions superficial.
D/E	0-0.4	Submit answer success not meaningfully tested.

Submit Answer - Not Found Cases (1 mark)

Band	Marks	Criteria
A	0.8-1	Both sub-cases tested separately: game session not found (repository returns <code>null</code> , asserts <code>404</code>) and question not found (repository returns <code>null</code> , asserts <code>404</code>).
B	0.7	Both sub-cases present but may be combined into one test or one assertion may be missing.
C	0.5-0.6	One sub-case tested.
D/E	0-0.4	Not-found cases not tested.

JWT Middleware Tests (2 marks)

Band	Marks	Criteria
A	1.6-2	All four cases tested. Missing header returns <code>401</code> and does not call <code>next</code> . Malformed header returns <code>401</code> . Valid token calls <code>next()</code> with no arguments and sets <code>req.user</code> with the decoded payload - token signed with <code>jwt.sign()</code> rather than mocked. Expired or tampered token returns <code>401</code> and does not call <code>next</code> . <code>afterEach(() => sinon.restore())</code> present.
B	1.3-1.5	All four cases present with minor omissions. <code>req.user</code> assertion may be missing for the valid token case. Token signing approach correct.
C	1-1.2	Two or three cases tested. Some assertions missing. JWT library may be incorrectly mocked rather than using a real signed token.
D/E	0-0.9	Fewer than two cases tested or middleware not meaningfully verified.

RBAC Middleware Tests (2 marks)

Band	Marks	Criteria
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Band	Marks	Criteria
A	1.6-2	All four cases tested. Undefined <code>req.user</code> returns <code>403</code> and does not call <code>next</code> . Unauthorised role returns <code>403</code> . Authorised role calls <code>next()</code> with no arguments. Multiple permitted roles configured - any matching role calls <code>next().afterEach(() => sinon.restore())</code> present.
B	1.3-1.5	All four cases present with minor omissions. Multiple-role case may only test one role from the list.
C	1-1.2	Two or three cases tested. Some assertions missing.
D/E	0-0.9	Fewer than two cases tested or middleware not meaningfully verified.

Code Coverage (2 marks)

Branch Coverage (1 mark)

Band	Marks	Criteria
A	0.8-1	At least 80% branch coverage achieved across all included files (<code>controllers/**/*.js</code> , <code>middleware/**/*.js</code>). <code>.c8rc</code> correctly configured with <code>"all": true</code> . Any <code>/* c8 ignore next */</code> usage is minimal and accompanied by a comment justifying why the line cannot be tested.
B	0.7	Branch coverage between 70–79%. Configuration mostly correct. Minor gaps in branch coverage traceable to untested error paths.
C	0.5-0.6	Branch coverage between 50–69%. Several error or conditional branches untested. <code>.c8rc</code> may have configuration issues.
D/E	0-0.4	Branch coverage below 50% or coverage tooling not configured or not running correctly.

Statement and Function Coverage (1 mark)

Band	Marks	Criteria
A	0.8-1	At least 80% statement and function coverage across all included files. Coverage thresholds set in <code>.c8rc</code> . <code>npm run test:coverage</code> runs without threshold failures.
B	0.7	Statement and function coverage between 70–79%. Thresholds set but one may fall slightly below 80%.
C	0.5-0.6	Coverage between 50–69%. Thresholds may not be configured. Some files have very low coverage.
D/E	0-0.4	Coverage below 50% or tooling not functional.